SOFTWARE ENGINEERING LAB

Experiment No. 4: Project-Based Data Flow Diagram (DFD)

Use your own selected software project

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1. Aim

To prepare a Context Diagram, Level 0 DFD and one Level 1 DFD for your own software project.

2. Important Instruction

This Lab 4 is based on Shield AI. All data flows and diagrams below describe the proposed scam-detection system.

3. Your Project Details

Project Title	Shield AI — Scam Detection System
One-line purpose of the project	Analyses user-submitted email/SMS/other text to identify potential scam signals and return a risk verdict with reasons.
Main users/external systems	End user; URL/domain reputation service (optional); system maintainer.
Three to six major functions	Accept text; preprocess; detect scam signals; check URLs; display verdict and explanation; optionally save history/feedback.
Earlier project document used (requirements/use case, if available)	Shield AI SRS — Experiment 1; requirements FR-01 to FR-08.

4. Before Drawing the DFD

Look at your project requirements/use cases. Write only the items that really belong to your project.

S. No.	External Entity	Data sent to system	Data received from system
1	End user	Message text / optional feedback	Risk verdict, risk explanation, lookup status
2	URL reputation service	—	URL/domain lookup request
3	System maintainer	Detection rules / configuration	Configuration status
4			
5			

5. Major Processes of Your Project

Write 3–6 major functions as processes. Use action words, for example: Manage Orders, Record Attendance, Generate Report.

Process No.	Process Name	What does this process do?
1.0	Capture Text	Accept and validate the text submitted by the user.
2.0	Preprocess Text	Normalise text and extract relevant features and URLs.
3.0	Analyse Scam Signals	Evaluate suspicious phrases, requests, urgency and URL signals.
4.0	Generate Result	Combine signals and present a risk verdict and explanation.
5.0	Record Scan / Feedback	Optionally store scan metadata and feedback.
6.0	Manage Configuration	Load authorised detection rules and settings.

6. Data Stores

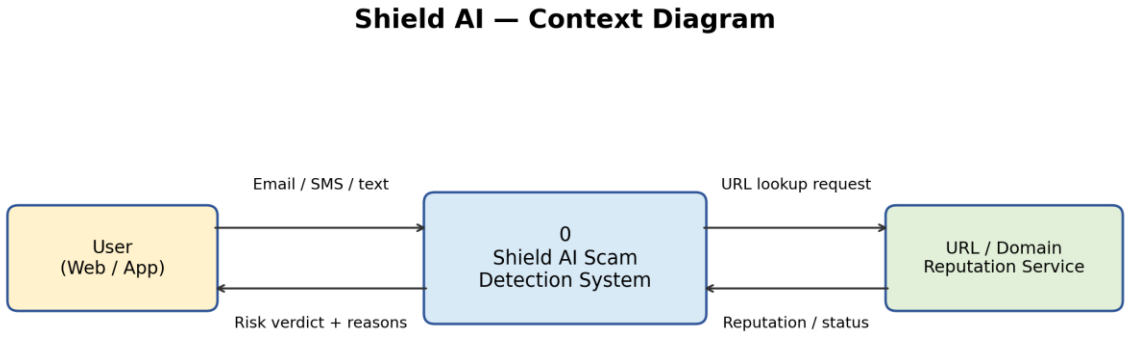
Write the important places where your project stores data.

Data Store	Name	What data is stored?
D1	Scan History	Scan time, input reference/metadata, verdict and risk signals; avoid storing full text unless explicitly required.
D2	Detection Rules / Model Config	Configured patterns, thresholds and model settings.
D3	Feedback Records	Optional user feedback associated with a scan ID.
D4	Service Configuration	Endpoint/settings for external URL reputation service, if configured.

7. Context Diagram

Draw the whole project as ONE process. Show the main external entities and major input/output data. Do not show internal processes or data stores here.

Paste/screenshot your Context Diagram below:

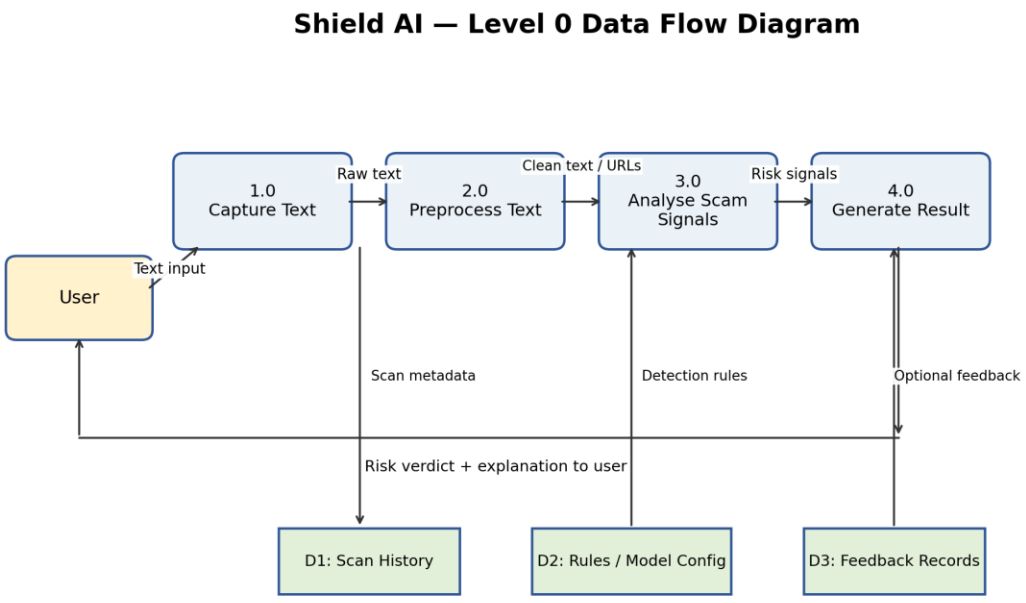


Check: Does the process contain your own project name? Yes

8. Level 0 DFD

Break your project into its major processes. Add required data stores and connect everything using labelled data arrows.

Paste/screenshot your Level 0 DFD below:

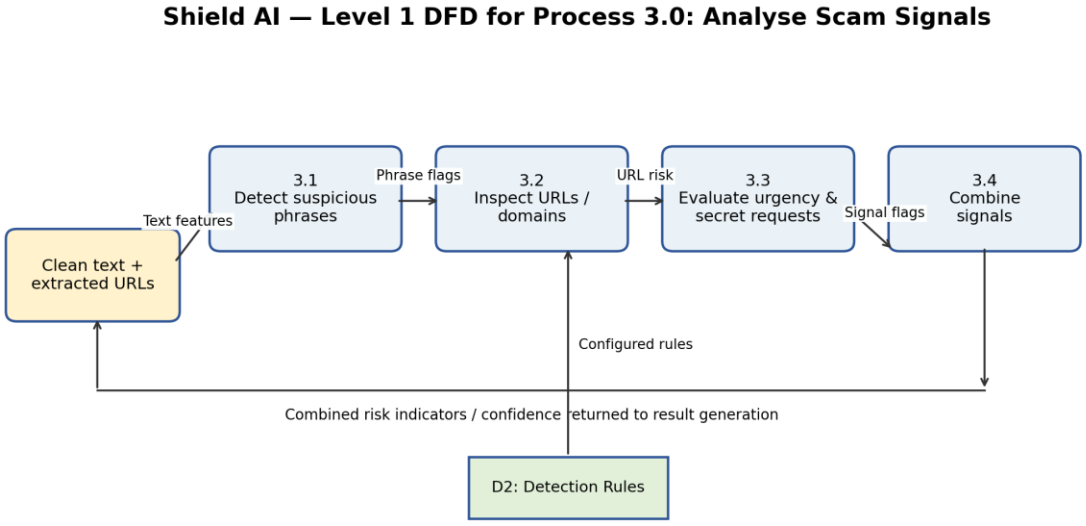


9. Level 1 DFD

Choose ONE process from your Level 0 DFD and break it into smaller steps.

Selected Level 0 Process	
3.0 Analyse Scam Signals	
It is the core decision process and combines message content and URL-related evidence before generating a risk assessment.	

Paste/screenshot your Level 1 DFD below:



10. Balancing Check

Check that the same important input/output data is represented consistently when you move from one DFD level to the next.

Check	Yes/No	Correction made, if any
Context Diagram and Level 0 are consistent.	Yes	User text enters the system and verdict/explanation returns to the user in both levels.
Selected Level 0 process and Level 1 are consistent.	Yes	Level 1 decomposes process 3.0 into signal-analysis steps and returns risk indicators.
No unrelated data suddenly appears in Level 1.	Yes	Level 1 uses only clean text, extracted URLs and configured detection rules.

11. Final Review

□ The work is based on my own project.

□ Process names match my project requirements/functions.

□ Arrows show data, not actions/control flow.

□ Every important arrow has a clear data label.

□ External entities are not directly connected to data stores.

□ Context, Level 0 and Level 1 are easy to read.

□ I can explain the diagram in simple words.

12. Result

The DFDs model Shield AI using a context diagram, a Level 0 breakdown and a Level 1 decomposition of scam-signal analysis. The diagrams show the main data exchanges, processes and data stores needed to produce a risk verdict.